



How cold was the Ice Age?

Researchers found out the temperature of the last ice age of 20,000 years ago - about 8 degrees Celsius. <http://digit.in/sep20-77>



Estrogen vs COVID-19

Study suggests that Estrogen may have a role to play in the symptoms being less severe in women. <http://digit.in/sep20-78>

In the mid 60s, astronomers studying a binary pulsar from the Arecibo Radio Observatory in Puerto Rico discovered a binary pair of neutron stars 21,000 light years away from the Earth. Two scientists, Russell A Hulse and Joseph H Taylor Jr studied the binary pair for 8 years, and in 1974, published their results. The pair of pulsars were getting closer together, their orbits decaying at the rate predicted by General Relativity. The loss in energy was exactly what was expected by the release of gravitational waves. There was now proof that the gravitational waves predicted by Einstein actually existed. Observations of other binary systems confirmed the findings. For this discovery, the two scientists shared the Nobel Prize for Physics in 1993.

In the 1960s, Joseph Weber used large cylinders of aluminum as gravitational wave detectors, hoping to detect deformations in the cylinders. Weber announced that he might have detected gravitational waves, but the findings were controversial, but he was one of the earliest scientists to actually try and directly observe gravitational waves, and since then he has been acknowledged for kickstarting the field of gravitational wave detection. The approach currently used by LIGO was suggested by Weber and another scientist called Rainer Weiss. Two Russian scientists, Michael Gertsenshtein and Vladislav Pustovoit independently suggested the same approach in 1962. Rainer Weiss went on to build many prototypes for gravitational wave detectors based on the same basic principle. The design proposed by Weiss is what is essentially used in the gravitational wave detectors of today.

The gravitational waves that can possibly be detected would have to be produced by massive objects, such as supernovae, black holes, quasars or pulsars. For some of the objects that are too dense to even let light escape, such as black holes, gravitational waves are the only way of detection and observation. Gravitational waves

are a rich source of information about these exotic objects that are still not very well understood. A gravitational wave detector can be made of anything, as long as the deformations of that thing can be accurately read. If a circle of test masses is suspended, then a gravitational wave passing through the circle will turn it into an oval, as the geometry of spacetime itself changes.

In 1976 Kip Thorne started a working group towards building dedicated instruments for detecting gravitational waves. In 1984, LIGO was established as a collaboration between two of the top technology universities in the United States, MIT and Caltech. Construction on LIGO began in 1992,



Joseph Weber with his cylindrical detector

after a few larger prototypes that failed to detect any waves. Analysis of the data was a massive undertaking, and over the years the collaboration grew to include more than a 1000 scientists from 15 universities around the world.

When a gravitational wave observatory such as LIGO or VIRGO detects a wave, the entire instrument is measuring itself deforming. These instruments work by using a laser beam split into two, that are then reflected back by mirrors and then made to produce an interference pat-

tern. Changes in the pattern indicates the passage of a gravitational wave. For a more sensitive instrument, you just need a longer distance for each of the arms of the detector, the distance over which the laser is directed before being bounced back by a mirror. A detector with longer arms is more sensitive. The VIRGO instrument has two 3 kilometer arms, while each of the two LIGO observatories have two 4 kilometer long arms.

In 2015, researchers at LIGO detected gravitational waves for the first time. They waited for a year to double check the data and make sure it was accurate, and published their findings in 2016. The gravitational waves from a collision between two black holes, at a distance of 1.3 billion light years. Just before the collision, the black holes, each 30 times as massive as the Sun were circling each other at speeds close to that of light. In 2017, VIRGO went online and joined the collaboration. Since then the detectors have observed colliding neutron stars, mergers between black holes of unequal masses, and many black hole mergers. The instruments allow us to see the universe in a whole new way, and the 2017 Nobel Prize in Physics was awarded to Rainer Weiss, Barry C Barish and Kip Thorne for their work on the LIGO/VIRGO collaboration.

Gravitational wave astronomy has just started off, and promises to play a major role in the future, particularly in the new era of multi messenger astronomy where the same targets are observed simultaneously by a number of instruments. Then in 2019, a fourth major gravitational wave observatory, set up in an abandoned mine in Japan, known as the Kamioka Gravitational Wave Detector (KAGRA) began operations. There are plans to build another detector, the third LIGO instrument, near Hingoli in Maharashtra. 